

ARTICLE INFORMATION

Article title

A Multi-Constituency Open Municipal Dataset of Fiscal Allocations, Constituency Development Fund (CDF) Interventions, and Local Governance in Ndola City, Zambia

Authors

1. Joshua Mvula 2021424014
2. Jonathan Mwale 2021512967
3. Enoch Simfukwe 2022004264
4. Humphrey Chama 2022027051
5. Mitchell Ndonga 2021395111

Affiliations

Department of Computer Science, School of Natural Sciences, The University of Zambia, Great East Road Campus, P.O. Box 32379, Lusaka, Zambia.

Corresponding author's email address and Twitter handle

team2.csc4792@unza.zm / lighton.phiri@unza.zm

Keywords

Ndola City Council; Constituency Development Fund (CDF); Local Government Act No. 2 of 2019; Municipal Open Data; Public Financial Management; Zambia Decentralization Policy

Abstract

This article describes a curated, multi-source municipal open dataset acquired from the Ndola City Council (NCC), the local government authority for Ndola District, Copperbelt Province, Zambia. Spanning fiscal years 2024 to 2026, the dataset captures 3,540 standardized records encompassing Constituency Development Fund (CDF) community infrastructure projects, tertiary skills development bursaries, secondary school boarding bursaries, youth/women empowerment loans, and grants across four constituencies (Ndola Central, Bwana M'kubwa, Chifubu, and Kabushi). Alongside this, the dataset includes municipal Output-Based Budgets (OBB) and council committee governance resolutions. Raw PDF reports and HTML trackers were ingested via automated scraping pipelines and cleaned into pipe-delimited (|) tabular files to provide a high-granularity snapshot of decentralized public investments.

SPECIFICATION TABLE

Subject	COMPUTER SCIENCE
Specific subject area	Data Mining, Municipal Open Data Extraction, Decentralized Fiscal Tracking, and Local Governance Transparency.
Type of data	Table Dataset
How data were acquired	Automated Python web scraping scripts (requests, BeautifulSoup4) targeting the official Ndola City Council portal (https://www.ndolacouncil.gov.zm), followed by programmatic PDF tabular and textual parsing (pdfplumber, pypdf, regex engine).
Data format	Raw Cleaned Processed
Data source location	Institution: Ndola City Council City/Region: Ndola, Copperbelt Province Country: Republic of Zambia Portal URL: https://www.ndolacouncil.gov.zm
Data accessibility	Repository name: Kaggle Data identification number: db-unza26-csc4792-ndola_city_council Direct URL to data: https://www.kaggle.com/datasets/unza26csc4792team2/ndola-city-council-open-data
Related research article	None

VALUE OF THE DATA

- Provides high-granularity, project-level and beneficiary-level fiscal records for tracking decentralized public investments across four constituencies in Zambia's second-largest commercial city under the reformed Constituency Development Fund (CDF) framework.
- Enables economists, policy analysts, and data scientists to evaluate equitable resource distribution across educational bursaries, infrastructure development (health, education, water/sanitation), and youth/women micro-enterprise capitalization.
- Serves as a benchmark tabular dataset for building predictive models on project completion rates, grant vs. loan performance, resource allocation variance, and socio-economic vulnerability targeting.
- Offers an open-source codebase demonstrating resilient tabular data extraction from unstructured municipal government PDFs and dynamic web portals in Sub-Saharan Africa, which can be reused for other Zambian councils.

BACKGROUND

The compilation of this dataset was motivated by the enactment of the Local Government (Amendment) Act No. 28 of 2023 and Act No. 76 of 2026, which restructured the Local Government Equalisation Fund (LGEF) and drastically increased the Constituency Development Fund (CDF) allocations to K40 million per constituency. Ndola City Council plays a pivotal role in decentralized service delivery across four distinct constituencies (Ndola Central, Bwana M’kubwa, Chifubu, and Kabushi). This dataset was generated to establish a structured, open-access repository of municipal fiscal allocations, bursary disbursements, community projects, and administrative resolutions. By converting unstructured statutory reports and web

trackers into standardized tabular formats, this dataset provides a much-needed baseline for empirical analysis, transparency tracking, and the evaluation of public financial management practices at the district level.

DATA DESCRIPTION

The curated dataset is structured into six standalone, pipe-separated (|) comma-separated value (.csv) files, conforming to the standardized naming convention `db-unza26-csc4792-[DESCRIPTION].csv`.

1. CDF Community Projects (`db-unza26-csc4792-ndola_council_cdf_community_projects_2024_2025.csv`)

Contains 216 records of approved and disapproved CDF infrastructure projects. Features include `project_id`, `project_name`, `project_description`, `sector` (e.g., EDUCATION, HEALTH), `project_type`, `constituency`, `ward`, `zone`, `location`, `amount_allocated_zmw` (ZMW), `approval_status`, `fiscal_year`, and `justification_notes`.

2. Skills Development Bursaries (`db-unza26-csc4792-ndola_council_cdf_skills_bursaries_2025.csv`)

Contains 2,476 beneficiary awards for tertiary, technical, and vocational skills. Features include `bursary_id`, `constituency`, `student_name`, `nrc_no`, `contact_no`, `ward`, `gender`, `age`, `institution_and_course`, `annual_fees_zmw`, `academic_year`, and `approval_status`.

3. Secondary School Boarding Bursaries (`db-unza26-csc4792-ndola_council_cdf_secondary_boarding_bursaries_2025.csv`)

Contains 77 awards to vulnerable pupils. Features include `secondary_bursary_id`, `constituency`, `pupil_name`, `ward`, `gender`, `age`, `grade_level`, `parent_or_guardian_name`, `school_name`, `school_location`, `annual_fees_zmw`, `academic_year`, and `approval_status`.

4. Empowerment Grants and Loans (`db-unza26-csc4792-ndola_council_cdf_empowerment_grants_loans_2025.csv`)

Contains 689 records of youth, women, and community cooperative disbursements. Features include `empowerment_id`, `constituency`, `funding_type` (GRANT/LOAN), `group_or_applicant_name`, `group_category`, `registration_no`, `ward`, `contact_person`, `business_venture_type`, `economic_sector`, `amount_approved_zmw`, `allocation_year`, and `approval_status`.

5. Municipal Budgets & Financials (`db-unza26-csc4792-ndola_council_municipal_budget_and_expenditure_2024_2026.csv`)

Contains 55 records of Output-Based Budget (OBB) estimates and revenue categories. Features include `budget_record_id`, `fiscal_year`, `programme_name`, `line_item_category`, `budget_type`, `amount_zmw`, `currency`, and `funding_source`.

6. Governance & Committee Resolutions (`db-unza26-csc4792-ndola_council_governance_and_committee_resolutions_2024_2025.csv`)

Contains 27 records of statutory committee resolutions. Features include `resolution_id`, `committee_or_body`, `meeting_date`, `agenda_item_or_ref`, `resolution_summary`, `governance_area`, and `status`.

EXPERIMENTAL DESIGN, MATERIALS AND METHODS

The data acquisition pipeline was engineered using Python, heavily relying on the `requests` and `BeautifulSoup4` libraries. A specialized web crawler systematically traversed the hierarchical website architecture of the Ndola City Council portal (<https://www.ndolacouncil.gov.zm>). The crawler navigated through departmental directories, public media archives, and the central CDF portal.

All static document links ending in `.pdf`, `.xlsx`, and `.docx` were cataloged into a JSON manifest. A dedicated downloader script retrieved 48 unique municipal source PDF artifacts (approximately 133 MB) using stream buffers and generated SHA-256 cryptographic hashes for forensic provenance verification.

Because municipal administrative records are published in heterogeneous PDF layouts (comprising both structured digital tables and tabular plain text with wrapped multi-line columns), a hybrid extraction strategy was deployed:

1. **Grid-Aligned Tables:** Extracted directly using the `pdfplumber` bounding-box table extraction algorithms, standardizing table header rows, removing nested subheaders, and interpolating empty merged cells.
2. **Text-Stream Parsing:** Deployed regular expression state machines to tokenize continuous text streams from PDF pages where table gridlines were absent. Expressions matched strict National Registration Card (NRC) patterns (`\d{6}/\d{2}/\d`), Zambian phone numbers (`\d{9,10}`), student age, gender tokens (`[MF]`), and terminal currency amounts.
3. **Data Cleaning & Normalization:**
 - Removed currency prefixes (ZMW, K, commas, spaces) and coerced all monetary fields into standard floating-point representations.
 - Standardized gender representations to single-character categorical tokens (M for Male, F for Female, U for Unknown).
 - Scanned all text features for internal pipe characters (`|`) and replaced them with forward slashes (`/`) to ensure strict CSV parsing compliance.
 - Generated alphanumeric primary keys (e.g., NCC-CDF-PRJ-XXXX) across all entity classes.

LIMITATIONS

The dataset is limited to statutory records and trackers that were publicly published on the Ndola City Council web portal. Specific project allocations or beneficiary records may be missing if the corresponding municipal documents were not uploaded online or were provided in heavily corrupted, non-machine-readable image formats (scans without OCR). Furthermore, certain demographic data points, such as the exact age and gender for specific bursary applicants, are registered as unknown (U or `Null`) where the source statutory reports contained missing entries.

ETHICS STATEMENT

The authors have read and follow the ethical requirements for publication in Data in Brief and confirm that the current work does not involve human subjects, animal experiments, or any data collected from social media platforms. All data collected and described in this article were obtained from public domain documents published by the Ndola City Council. No private non-public information was breached, and all personal beneficiary entries reflect official statutory public disclosures legally mandated under the Zambian CDF Act.

CREDIT AUTHOR STATEMENT

Project Team #2: Conceptualization, Methodology, Software, Data Curation, Investigation, Writing - Original Draft, Validation.

Lighton Phiri: Supervision, Project Administration, Quality Assurance.

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DECLARATION OF COMPETING INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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